

RESEARCH AREAS AND EXPERTISE

Theoretical and Computational Fluid Mechanics

I use numerical and analytical methods to study fluid flows in which heat, chemical species, and electric fields interact with complex surfaces such as porous media and selectively permeable membranes. My work is motivated by applications to desalination, CO₂ sequestration, laminar flow control, micro-robotics, and targeted drug delivery. My numerical work focuses on finite volume methods, immersed boundary methods, spectral methods, and projection methods. My analytical work focuses on perturbation methods, linear/nonlinear stability analyses, and volume-averaged models of flow through porous media.

EDUCATION

- Doctor of Philosophy**, Mechanical Engineering 2006 - 2009
McGill University, Montreal, Canada
Advisor: Dr. Luca Cortelezzi
Thesis: *The effects of wall permeability on the linear stability of channel flow and the asymptotic suction boundary layer*
- Master of Engineering**, Mechanical Engineering 2004 - 2006
McGill University, Montreal, Canada
Advisor: Dr. Luca Cortelezzi
Thesis: *A linear stability analysis of the destabilizing effects of wall-permeability in channel flows*
- Bachelor of Engineering**, Mechanical Engineering, Honors Program 1998 - 2003
McGill University, Montreal, Canada
Advisor: Dr. Paul Zsombor-Murray
Honors Thesis: *The kinematic analysis of a planar double triangular parallel manipulator*

PROFESSIONAL EXPERIENCE

- Associate Professor**, Colorado School of Mines, Golden CO 2021 - Present
- Assistant Professor**, Colorado School of Mines, Golden CO 2014 - 2021
- Visiting Professor**, Ecole Centrale de Marseille, France June 2015
- Postdoctoral Research Fellow**, University of Maryland, College Park 2011 - 2014
Advisor: Dr. Amir Riaz
Buoyancy driven convection during CO₂ sequestration.
- Postdoctoral Research Fellow**, University of Aix-Marseille, France 2009 - 2011
Advisors: Drs. Denis Martinand, Eric Serre, and Richard Lueptow
Numerical simulations of membrane filtration in Taylor-Couette cells.
- Graduate Research Assistant**, McGill University, Montreal, Canada 2003 - 2009
Advisor: Dr. Luca Cortelezzi
Effects of porous walls on stability of channel flows and boundary layers.

ARCHIVAL PUBLICATIONS (advisees underlined)

1. K. Koponen, A.P. Singh Bhalla, B. Sprinkle, N. Wu, N. Tilton, 2025 “A direct forcing, immersed boundary method for conjugate heat transport,” *Journal of Computational Physics*, in press. <https://doi.org/10.1016/j.jcp.2025.114135>
2. M. Soleimani, K. Koponen, N. Tilton, A.P.S. Bhalla 2026, “Analytical and numerical solutions to the three-phase Stefan problem with simultaneous occurrences of melting, solidification, boiling, and condensation phenomena,” *ASME Journal of Heat and Mass Transfer*, **148**, 148, 012401, <https://doi.org/10.1115/1.4069329>
3. S. Molins, N. Spycher, S. M. Dischinger, M. Nassr, J.Y. Lee, J. Johnston, N. Tilton, K. Gleason, B.D. Freeman, W.T. Stringfellow, D.J. Miller, 2025 “Permeate Fluxes from Desalination of Brines and Produced Waters: A Reactive Transport Modeling Study,” *Applied Geochemistry*, in press. <https://doi.org/10.1016/j.apgeochem.2025.106429>
4. M. Nassr, S.M. Dischinger, J.Y. Lee, K.L. Gleason, S. Molins, N. Spycher, P. Bigdelou, J. Johnston, N.A. Lynd, N. Tilton, W.T. Stringfellow, B.D. Freeman, D.J. Miller, 2025 “Mineral scale formation during crossflow reverse osmosis at constant flux and constant transmembrane pressure conditions,” *Industrial & Engineering Chemistry Research* **64**, 1295–1308. <https://doi.org/10.1021/acs.iecr.4c04059>
5. F. Municchi, Y. Liu, J. Wang, T.Y. Cath, C.S. Turchi, M.B. Heeley, E.M.V. Hoek, D. Jassby, N. Tilton, 2024 “A numerical method for simulating variable density flows in membrane desalination systems,” *Computers & Fluids*, **285**, 106449. <https://doi.org/10.1016/j.compfluid.2024.106449>
6. M. Mabry, F. Municchi, Y. Liu, J. Wang, T.Cath, C.Turchi, M. Heeley, E.M.V. Hoek, D. Jassby, D. Martinand, N. Tilton, 2024 “Generating buoyancy-driven convection in membrane distillation,” *Journal of Membrane Science*, **709**, 123043. <https://doi.org/10.1016/j.memsci.2024.123043>
7. M. Heeley, Y. Liu, J. Municchi, J. Lou, M. Dudley, M. Kaddoura, E.M.V Hoek, N. Tilton, C, Turchi, T. Cath, D. Jassby, 2024 “Economic assessment of a multistage surface-heated vacuum membrane distillation process for the treatment of hypersaline produced water,” *Desalination*, **582**, 117648. <https://doi.org/10.1016/j.desal.2024.117648>
8. J. Johnston, S.M. Dischinger, M. Nassr, J.Y. Lee, P. Bigdelou, B.D. Freeman, K.L. Gleason, D. Martinand, D.J. Miller, S. Molins, N. Spycher, W.T. Stringfellow & N. Tilton, 2023 “A reduced-order model of concentration polarization in reverse osmosis systems with feed spacers,” *Journal of Membrane Science*, **675**, 121508. <https://doi.org/10.1016/j.memsci.2023.121508>
9. J. Lou, M. Dudley, J. Wang, Y. Liu, T.Y. Cath, C.S. Turchi, M.B. Heeley, E.M.V. Hoek, D. Jassby & N. Tilton, 2023 “Numerical simulations of membrane distillation systems with actively heated membranes,” *Journal of Membrane Science*, **668**, 121206. <https://doi.org/10.1016/j.memsci.2022.121206>
10. Y. Liu, J. Wang, B. Jung, U. Rao, E. Sedighi, E.M.V. Hoek, N. Tilton, T.Y. Cath, C.S. Turchi, M.B. Heeley, Y.S. Ju & D. Jassby, 2022 “Desalinating a real hyper-saline pretreated produced water via direct-heat vacuum membrane distillation,” *Water Research*, **218**, 118503. <https://doi.org/10.1016/j.watres.2022.118503>
11. Y. Liu, J. Wang, E.M.V. Hoek, F. Municchi, N. Tilton, T.Y. Cath, C.S. Turchi, M.B. Heeley & D. Jassby, 2022 “Multistage surface-heated vacuum membrane distillation process enables high water recovery and excellent heat utilization: A modeling study,” *Environmental Science & Technology*, **57**, 643-654. <https://doi.org/10.1021/acs.est.2c07094>
12. J. Johnston, J. Lou & N. Tilton, 2022 “Application of projection methods to simulating mass transport in reverse osmosis systems,” *Computers & Fluids*, **232**, 105189 <https://doi.org/10.1016/j.compfluid.2021.105189>
13. R. Ben Dhia, N. Tilton & D. Martinand, 2022 “Impact of osmotic pressure on the stability of Taylor vortices,” *Journal of Fluid Mechanics*, **933**, 1-33, <https://doi.org/10.1017/jfm.2021.1101>
14. J. Wang, Y. Liu, U. Rao, M. Dudley, N.D. Ebrahimi, E.M.V. Hoek, N. Tilton, T. Y. Cath, C.S. Turchi, Y.S. Jub & D. Jassby, 2021 “Conducting solar energy to the membrane/water interface for the enhanced desalination of hypersaline brines using membrane distillation,” *Journal of Membrane Science*, **626**, 119188, <https://doi.org/10.1016/j.memsci.2021.119188>
15. J. Lou, J. Johnston, T.Y. Cath, D. Martinand & N. Tilton, 2021 “Computational fluid dynamics simulations of unsteady mixing in spacer-filled direct contact membrane distillation channels,” *Journal of Membrane Science*, **633**, 118931, <https://doi.org/10.1016/j.memsci.2020.118931>

16. Y. Guo, J. Lou, J. K. Cho, N. Tilton, J. Chun, W. Um, X. Yin, K. Neeves & N. Wu, 2020 “Transport of colloidal particles in microscopic porous medium analogs with surface charge heterogeneity: experiments and fundamental role of single-bead deposition,” *Environmental Science & Technology*, 54, 21, 13651–13660, <https://doi.org/10.1021/acs.est.0c03225>
17. J. Lou, J. Johnston & N. Tilton, 2020 “Application of projection and immersed boundary methods to simulating heat and mass transport in membrane distillation,” *Computers & Fluids*, 212, 104711, <https://doi.org/10.1016/j.compfluid.2020.104711>
18. Z. Khalifa, L. Pocher & N. Tilton, 2020 “Regimes of flow through cylinder arrays subject to steady pressure gradients,” *International Journal of Heat and Mass Transfer*, 159, 120072, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120072>
19. D. Parikh, Y. Wu, C. Peterson, S. Jarriel, M. Mooney & N. Tilton, 2019 “The coupled dynamics of foam generation and pipe flow,” *International Journal of Heat and Fluid Flow*, 79, 108442, <https://doi.org/10.1016/j.ijheatfluidflow.2019.108442>
20. J. Lou, J. Vanneste, S.C. Decaluwe, T.Y. Cath & N. Tilton, 2019 “Computational fluid dynamics simulations of polarization phenomena in direct contact membrane distillation,” *Journal of Membrane Science*, 588, 117150, <https://doi.org/10.1016/j.memsci.2019.05.074>
21. N. Tilton & D. Martinand, 2018 “Taylor-Couette-Poiseuille flow with a weakly permeable inner cylinder: absolute instabilities and selection of global modes,” *Journal of Fluid Mechanics*, 849, 741–776, <https://doi.org/10.1017/jfm.2018.437>
22. N. Tilton, 2018 “Onset of transient natural convection in porous media due to porosity perturbations,” *Journal of Fluid Mechanics*, 838, 129–147, <https://doi.org/10.1017/jfm.2017.861>
23. N. Tilton & L. Cortelezzi, 2015 “Stability of boundary layers over porous walls with suction,” *AIAA Journal*, 53, 2856–2868, <https://doi.org/10.2514/1.J053716>
24. M.A. Nomeli, N. Tilton & A. Riaz, 2014 “A new model for the density of saturated solutions of CO₂-H₂O-NaCl in saline aquifers,” *International Journal of Greenhouse Gas Control*, 31, 192–204, <https://doi.org/10.1016/j.ijggc.2014.10.006>
25. N. Tilton & A. Riaz, 2014 “Nonlinear stability of gravitationally unstable transient boundary layers in porous media,” *Journal of Fluid Mechanics*, 745, 251–278, <https://doi.org/10.1017/jfm.2014.72>
26. N. Tilton, E. Serre, D. Martinand & R.M. Lueptow, 2014 “A 3D pseudospectral algorithm for fluid flows with permeable walls: application to filtration,” *Computers & Fluids*, 93, 129–145, <https://doi.org/10.1016/j.compfluid.2014.01.003>
27. N. Tilton, D. Daniel & A. Riaz, 2013 “The initial transient period of gravitationally-unstable diffusive boundary layers developing in porous media,” *Physics of Fluids*, 25, 092107, <https://doi.org/10.1063/1.4821225>
28. D. Daniel, N. Tilton & A. Riaz, 2013 “Optimal perturbations of gravitationally unstable, transient boundary layers in porous media,” *Journal of Fluid Mechanics*, **727**, 456–486, <https://doi.org/10.1017/jfm.2013.277>
29. N. Tilton, D. Martinand, E. Serre & R.M. Lueptow, 2012 “Incorporating Darcy’s law for pure solvent flow through porous tubes: asymptotic solution and numerical simulations,” *AIChE Journal*, **58**, 2030–2044, <https://doi.org/10.1002/aic.13823>
30. N. Tilton, D. Martinand, E. Serre & R.M. Lueptow, 2010 “Pressure-driven radial flow in a Taylor-Couette cell,” *Journal of Fluid Mechanics*, **660**, 527–537, <https://doi.org/10.1017/S0022112010003228>
31. N. Tilton & L. Cortelezzi, 2008 “Linear stability analysis of pressure driven flows in channels with porous walls,” *Journal of Fluid Mechanics*, **604**, 411–445, <https://doi.org/10.1017/S0022112008001341>
32. N. Tilton & L. Cortelezzi, 2006 “The destabilizing effects of wall-permeability in channel flows: A linear stability analysis,” *Physics of Fluids*, **18**, 051702, <https://doi.org/10.1063/1.2202649>

INVITED TALKS & SEMINARS

1. “Reduced Models of Concentration Polarization in RO Systems,” NAMS Invited Session of Water Innovation using Membranes, August 31, 2021.
2. “Numerical Simulations of Heat and Mass Transport in Membrane Filtration Systems,” University of Utah, March 8, 2021
3. “Numerical Simulations of Heat and Mass Transport in Membrane Filtration Systems” National Alliance for Water Innovation Seminar Series, October 5 , 2020
4. “Numerical Simulations of Heat and Mass Transport in Membrane Filtration Systems,” Mechanical and Aeronautical Engineering Research Seminar Series, University of Colorado, Colorado Springs, March 12, 2020
5. “Numerical Simulations of Heat and Mass Transport in Membrane Filtration Systems,” The Burgers Program Fluid Dynamics Reviews Seminar Series, University of Maryland, Sept 27, 2019.
6. “Direct Numerical Simulations of Unsteady Mixing and Concentration Polarization in Reverse Osmosis Systems,” 9th Sino-US Joint Conference of Chemical Engineering, Beijing, China, Oct 15-19, 2017”
7. “Numerical Simulations of Concentration Polarization in Reverse Osmosis Systems,” CEE Departmental Seminar, Colorado School of Mines, September, 2016
8. “Onset of Natural Convection During CO₂ Sequestration in Saline Aquifers,” Grand Challenges in Geological Fluid Mechanics, Los Alamos National Lab, Sept 2, 2016.
9. “Gravitational Instability and Convective Mixing in CO₂ Sequestration,” Ecole Central Marseille, France, June, 2015.
10. “Optimization and Perturbation Methods for Predicting Onset of Convection in CO₂ Sequestration,” Boulder Fluid Dynamics Seminar, CU Boulder, November, 2014.
11. “Optimization and Perturbation Methods for Predicting Onset of Convection in CO₂ Sequestration,” Applied Math & Statistics Seminar, Colorado School of Mines, September, 2014.

CONFERENCE PRESENTATIONS IN PAST 5 YEARS (advisees underlined)

1. K. Koponen, N. Wu, B. Sprinkle, A.P. Singh Bhalla, 2024 “An immersed boundary method for simulating the electric potential in flows with active colloids,” Annual meeting of the American Physical Society Division of Fluid Dynamics (APS-DFD), Nov 24-26, Salt Lake City, Utah.
2. K. Koponen, A.P.S Bhalla & N. Tilton, 2023 “A second-order, direct forcing, immersed boundary method for conjugate heat transport,” APS-DFD, Nov 19-21, Washington, DC.
3. N. Tilton & Federico Municchi, 2023 “Harnessing gravity to mitigate polarization and increase permeate production in membrane distillation,” Annual Meeting of North American Membrane Society (NAMS), May 13-17, Tuscaloosa, AL.
4. N. Tilton & J. Johnston, 2023 “A new approach to simulating concentration polarization in reverse osmosis systems with feed spacers,” Annual Meeting of NAMS, May 13-17, Tuscaloosa, AL.
5. N. Tilton, J. Johnston, S. Dischinger, M. Nassr, J.Y. Lee, B. Freeman, K. Gleason, D. Miller, S. Molins, N. Spycher, W. Stringfellow, 2022 “A reduced-order model of concentration boundary layers in reverse osmosis systems with vortical flow structures,” APS-DFD, Nov 20-22, Indianapolis, IN.
6. F. Municchi, J. Wang, M. Mabry, Y. Liu, T.Y. Cath, C. Turchi, M. Heeley, E.M.V. Hoek, D. Jassby & N. Tilton, 2022 “Harnessing buoyancy-driven instability to enhance thermal membrane desalination,” APS-DFD, Nov 20-22, Indianapolis, IN.
7. M. Dudley, N. Tilton, D. Jassby, E.M.V Hoek, C. Turchi, T.Y. Cath & M. Heeley, 2020 “3D Numerical simulations of temperature polarization in vacuum membrane distillation systems with active membrane heating,” NAMS, May 18-21, Tempe, AZ.

8. J. Lou, J. Johnston, D. Martinand & N. Tilton, 2020 “Direct numerical simulations of unsteady mixing in direct contact membrane distillation systems with different spacer blockages and vertical offsets,” NAMS, May 18-21, Tempe, AZ.
9. J. Johnston, J. Lou & N. Tilton, 2019 “A Direct Forcing Immersed Boundary Method for Simulations of Heat and Mass Transport with Neumann Boundary Conditions,” APS-DFD, Nov 23-26, Seattle, WA.
10. D. Martinand & N. Tilton, 2019 “Global modes in Taylor–Couette–Poiseuille Flow with a permeable inner cylinder,” APS-DFD, Nov 23-26, Seattle, WA.
11. J. Lou, J. Johnston & N. Tilton, 2019 “Transition to vortex shedding and its impact on heat and solute transport in membrane filtration systems,” APS-DFD, Nov 23-26, Seattle, WA.
12. J. Lou, J. Johnston & N. Tilton, 2019 “Direct numerical simulation of unsteady mixing in direct contact membrane distillation systems with membrane spacers,” NAMS, May 11-15, Pittsburgh, PA.
13. J. Johnston, J. Lou & N. Tilton, 2019 “Direct Numerical Simulation of Unsteady Mixing in Direct Contact Membrane Distillation Systems with Membrane Spacers” RMFM, July 29, Boulder CO.
14. M. Dudley & N. Tilton, 2019 “Numerical Study of Membrane Heating in a Vacuum Membrane Distillation System,” Rocky Mountain Fluid Mechanics Symposium (RMFM), July 29, Boulder CO.

FUNDED RESEARCH GRANTS (\$2.7 MILLION USD)

My role as Principle Investigator (PI) or co-PI is indicated.

1. U.S. National Science Foundation, **\$408,664** (PI) “Harnessing buoyancy-driven instabilities for enhanced membrane desalination,” 2023-2026.
2. National Alliance for Water Innovation, **\$227,202** (PI) “Computational Test Bed for Predictive Fouling Control,” 2020-2021
3. Colorado Office of Economic Development and International Trade. **\$35,000** (PI) “Prototyping smart spacers for improved water treatment,” 2020-2022.
4. Colorado Department of Higher Education, **\$7000** (PI) “Developing an OER textbook for a core graduate course in fluid mechanics,” 2020-2021.
5. U.S. Department of Energy, **\$818,927** (co-PI), “Energy where it matters: Delivering heat to the membrane/water interface for enhanced thermal desalination,” 2018-2023.
6. National Science Foundation Career Award, **\$547,346** (PI), “Robust numerical modeling for rational design of membrane filtration processes,” 2018-2023.
7. French Embassy, Thomas Jefferson Fund, **\$10,000** (PI) “Numerical and theoretical models for optimal mixing in membrane desalination systems.” Funded travel for my students to the University of Aix-Marseille between 2018-2022.
8. Colorado Office of Economic Development and International Trade, **\$150,000** (PI), “High-efficiency membrane desalination systems using new 3D-printed spacers for enhanced turbulence,” 2017-2019.
9. Newmont Mining Academic Associates Program, **\$30,000** (PI), 2017-2020.
10. U.S. Bureau of Reclamation, **\$143,869** (PI), “Overcoming the Technical Barriers to Membrane Distillation: Conductive Heat Losses and Polarization Phenomena,” 2016-2017.
11. Slater Family Research Fund, **\$169,544** (PI), “Membrane Distillation: Advanced Desalination and Water Treatment,” 2015-2018.

ADVISING

Post-Docs

1. Pedram Bigdelou
2. Jincheng Lou
3. Federico Municchi

PhD

1. Zahra Khalifa
2. Jincheng Lou
3. Jacob Johnston
4. Rouae Ben Dhia
5. Kimmo Kopponen

MS Thesis

1. Spencer Gilleon
2. Dhruvad Parikh
3. Mark Dudley
4. Nick Yearout
5. Miles Mabry

Undergraduate Research Assistants: (20 total) Benjamin Gallman, Stephanie Ng, Scott Jarriel, Ray Karam, Liam Pocher, Christian Peterson, Kody Von Holdt, Noah Pelsmaeker, Griffin Stanton, April Wood, Mitchell White, Alec Mullen, Jewel Newman, Cameron Howard, Noah Ford, Luke Scott, Alex Fong, Garrett Macket, Kaden Armstrong, Madeline Travis.

TEACHING

1. MEGN 551 (graduate), Advanced Fluid Mechanics, 2015-present
2. MEGN 502 (graduate), Advanced Engineering Analysis, 2019-present
3. MEGN 651 (graduate), Advanced Computational Fluid Dynamics, 2018-present
4. MEGN 361 (undergraduate), Thermodynamics 1, 2024
5. MEGN 351 (undergraduate), Fluid Mechanics, 2014-2018

PROPOSAL REVIEW PANELS

NSF Chemical and Biological Separations, NSF Process Separations, NSF Fluid Dynamics, NSF Career Panel, DOE, Israel Science Foundation, Canadian Natural Sciences and Engineering Research Council

JOURNAL REVIEWER

Journal of Fluid Mechanics, Physics of Fluids, Journal of Computational Physics, Computers & Fluids, International Journal of Heat and Mass Transfer, Journal of Membrane Science, Transport in Porous Media, Journal of Porous Media, Journal of Chemical Physics, Journal of Fluids, Physical Review Fluids, AIChE Journal, Revista Mexicana de Física, Fluid Dynamics Research, Chem Eng Commun, Trans Can Soc Mech Eng, SIAM Journal on Applied Mathematics, Water Resources Research, The Royal Society Proceedings A, Geophysical Research Letters, Desalination.